Scalling Laws for Native Multimodal Models

Dimas Christos

Electrical and Computer Engineering
Technical University of Crete

February 12, 2026

Contents

1. Introduction & Motivation

2. Related Work

3. Technical Approach, Experiments & Results

4. Limitations & Conclusion

Introduction & Motivation

The Goal

The Vision

Building general-purpose models that perceive the world through both **text** and **images** (and beyond) as naturally as humans do.

Current Standard (Late-Fusion):

- "Glue" a pre-trained Vision Encoder (e.g., CLIP) to an LLM.
- Treat modalities separately until late stages.

The Challenger (Early-Fusion):

- Train from scratch on raw inputs.
- Treat images as just another language (tokens).

Research Question: Is the "glued" approach actually better, or can we train efficient Native Multimodal Models (NMMs) from scratch?

A sneak pick on the findings

Performing an extensive scalling laws study there is surprisingly proven that:

- There is not an inheritent advantage to Late-fusion architectures over Early-fusion ones. They perform on par, with Late-fusion requiring more parameters.
- NMMs scale similarly to LLMs.
- Using Early-fusion architectures with Mixture of Experts (MoE) leads to more efficient models that better handle the heterogeneity of multimodal data, since it adds some specialization.
- Modality-agnositic routing beats modality-aware for sparse NMMs

Related Work

Context & Background

- **LLM Scaling Laws:**
  - Kaplan et al. and Chinchilla (Hoffmann et al.) established that performance scales predictably with compute (C), parameters (N), and data (D).
- **Multimodal Architectures:**
  - *Late-Fusion:* CLIP, Flamingo, LLaVA (Pre-trained backbones).
  - *Early-Fusion:* Chameleon, Fuyu (Tokenized images, less explored).
- **Mixture of Experts (MoE):**
  - Sparsely activated models (e.g., Mixtral) save compute.
  - This paper explores MoE to handle the *heterogeneity* of text vs. image data.

Definitions Part 1

To better explanation of this work, we will introduce here some of the main concepts that are key to understand the paper results:

Native Multimodal Models

Models trained from scratch on all modalities simultaneously, without relying on pre-trained LLM & Vision Encoder.

Early-Fusion Architectures

Enable multimodal interaction from the beggining, using a single transformer model, process raw multimodal input. Consider the main transformer as the Decoder.

Late-Fusion Architectures

Delaying multimodal interaction, typically after each separate unimodal component process each modality independently. Consider a Vision Encoder connected to a Decoder.

Definitions Part 2

Mixture of Experts (MoE)

Enables model to dynamically allocate specialized parameters across modalities in a symmetric and parallel manner, while late-fusion is assymetric and sequential (vision encoder first, then decoder). Scalling the number of training tokens is more important than the number of active parameters here.

Routing

In the context of MoE and Multimodal models, routing is the process of deciding which specialized sub-network (expert) should handle the incoming token.

Definitions Part 3

Modality-agnositic routing

In sparse MoE models, it refers to relying on a learned router module, trained jointly with the model.

Modality-aware routing

Routing based on pre-defined rules such as modality type routing (e.g. vision-tokens, text-tokens)

Technical Approach, Experiments & Results

Experiment Setup

- In early-fusion models, image patches are linearly projected to match the text token dimension, while late-fusion follows the CLIP architecture
- Training is conducted on a mixture of public and private multimodal datasets, including DCLM, Obelics, DFN, COYO, and a private collection of High-Quality Image-Text Pairs (HQITP). Images are resized to 224×224 resolution with a 14×14 patch size. A context length of 1k is used for the multimodal sequences. For training efficiency, models are trained with bfloat16, Fully Sharded Data Parallel (FSDP), activation checkpointing, and gradient accumulation. We also use sequence packing for the image captioning dataset to reduce the amount of padded tokens. Performance is evaluated on held-out subsets of interleaved (Obelics), Image-caption (HQITP), and text-only data (DCLM).

Deriving Scaling Laws for NMMs

Trained **457 models** across varying compute budgets to fit the power-law equation:

$$L(N, D) = E + \frac{A}{N^{\alpha}} + \frac{B}{D^{\beta}} \tag{1}$$

- L: Validation Loss (lower is better)
- E: Baseline Loss (irreducible error)
- N: Model Parameters
- D: Training Tokens
- α, β: Scaling exponents (how fast performance improves)

Goal: Find the "Compute-Optimal" frontier—the perfect balance of N and D for a given budget, for Early vs. Late Fusion, in comparison to LLMs and using different data mixtures.

Drive towards Multimodal Specialization

This step is going to be achieved by applying the Mixture of Experts (MoE) technique, to highlight the advantages of sparse early- fusion models over their dense counterparts.

Sparsity

Instead of a dense model, use a **Router** to select the top-k experts for each token.

Why Multimodal Specialization?

- Images and Text have different statistical properties.
- Hypothesis: Experts will naturally specialize (e.g., "Image Experts" vs. "Text Experts").
- Allows scaling parameters (Capacity) without exploding inference cost (FLOPs).

Goal: Optimize model's behavior and determine the proper routing strategy (modality-agnostic vs. modality-aware) for handling the heterogeneity of multimodal data.

Evaluate on downstream tasks with SFT

We conduct a multimodal instruction tuning stage (SFT) on the LLaVA mixture and report accuracy and CIDEr scores across several VQA and captioning tasks.

Goal: Following previous work on scaling laws, we primarily rely on validation losses. However, we need to generally check if this evaluation correlates well with performance on downstream tasks.

Result 1: No Inherent Advantage for Late Fusion

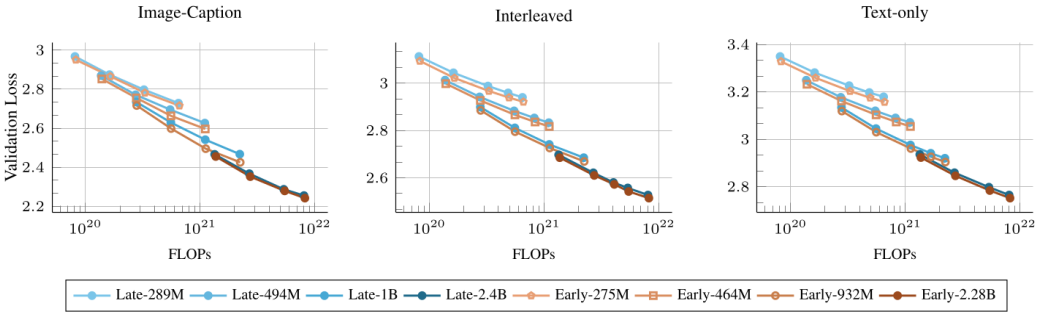
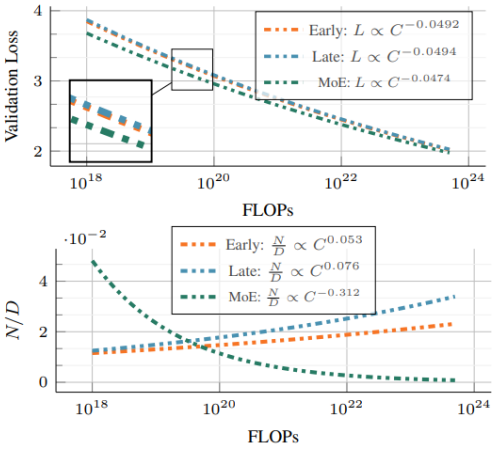


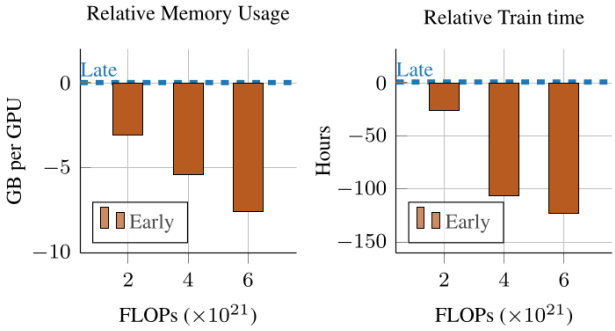
Figure: **Early vs late fusion:** scaling training FLOPs. We compare early and late fusion models when scaling both the number of model parameters and the number of training tokens.

- **Tie:** Both architectures perform similarly at scale.
- **Efficiency:** Early-fusion is better at lower budgets (fewer parameters, no heavy vision encoder).

Result 2: Efficiency & Trade-offs on Early vs Late Fusion



(a) **Scaling Properties**



(b) **TrainingEfficiency**

We observe that early-fusion is more efficient and faster to train at lower budgets needing more training tokens. Late-fusion can catch up at higher budgets needing more parameters.

Result 3: Scaling Laws for NMMs VS LLMs

$L = E + \frac{A}{N^\alpha} + \frac{B}{D^\beta}$		$N \propto C^a$		$D \propto C^b$	$L \propto C^c$		$D \propto N^d$	
Model	Data	E	α	β	a	b	c	d
GPT3 [10]	Text	–	–	–	–	–	-0.048	
Chinchilla [26]	Text	1.693	0.339	0.285	0.46	0.54	–	
NMM (early-fusion)	Text	2.222	0.3084	0.3375	0.5246	0.4774	-0.0420	0.9085
	Image-Caption	1.569	0.3111	0.3386	0.5203	0.4785	-0.0610	0.9187
	Interleaved	1.966	0.2971	0.338	0.5315	0.4680	-0.0459	0.8791
	AVG	1.904	0.301	0.335	0.5262	0.473	-0.0492	0.8987
NMM (late-fusion)	AVG	1.891	0.2903	0.3383	0.6358	0.4619	-0.0494	0.6732
Sparse NMM (early-fusion)	AVG	2.158	0.710	0.372	0.361	0.656	-0.047	1.797

- Comparing the scaling law coefficients of our NMMs to those reported for text-only LLMs (e.g., GPT-3, Chinchilla), we find them to be within similar ranges. In particular, for predicting the loss as a function of compute, GPT-3 follows L $\propto$ C - 0.048 , while our models follow L $\propto$ C - 0.049 , suggesting that the performance of NMMs adheres to similar scaling laws as LLMs.

Result 4: Data Mixture Matters

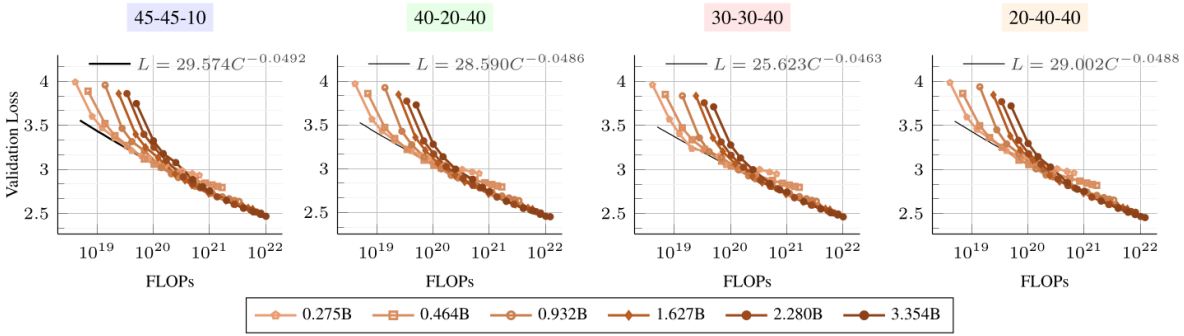
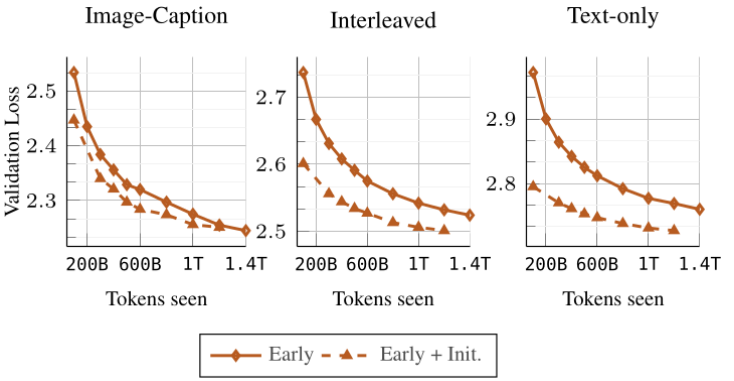


Figure: Scaling laws with different training mixtures. Early-fusion models follow similar scaling trends when changing the pretraining mixtures. However, increasing the image captions leads to a higher scaling exponent norm. Data mixture is in C-I-T form (image-caption, interleaved and text)

Result 5: Native multimodal pre-training vs continual training of LLMs



We can see that native multimodal models can close the gap with initialized models when trained for longer. Considering the cost of pre-training, these results suggest that training natively could be a more efficient approach for achieving the same performance on multimodal benchmarks.

Sparse models setup

Sparse models are based on the dropless-MoE implementation, which eliminates token dropping during training caused by expert capacity constraints.

A top-k expert-choice routing mechanism is employed, where each token selects its top-k experts among the E available experts. Specifically, it was set $k = 1$ and $E = 8$, as this was the most effecient configuration found.

Additionally, there is incorporated an auxiliary load-balancing loss with a weight of 0.01 to ensure a balanced expert utilization.

Result 6: Sparsity (MoE) Wins

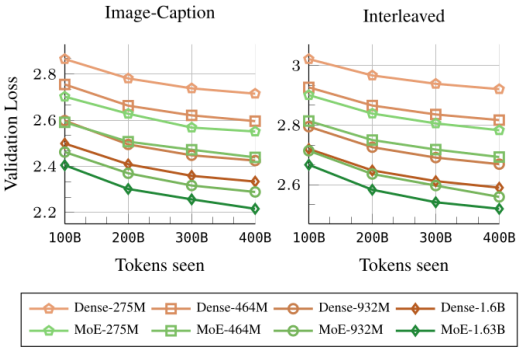


Figure: MoE vs Dense: Scaling training FLOPs.

Under the same inference cost (or number of active parameters),MoE significantly outperform dense models. Interestingly, this performance gap is more pronounced for smaller model sizes. This suggests that MoE enable models to handle heterogeneous data more effectively and specialize in different modalities. However, as dense models become sufficiently large, the gap between the two architectures gradually closes.

Result 7: Scaling Laws for Sparse NMMs

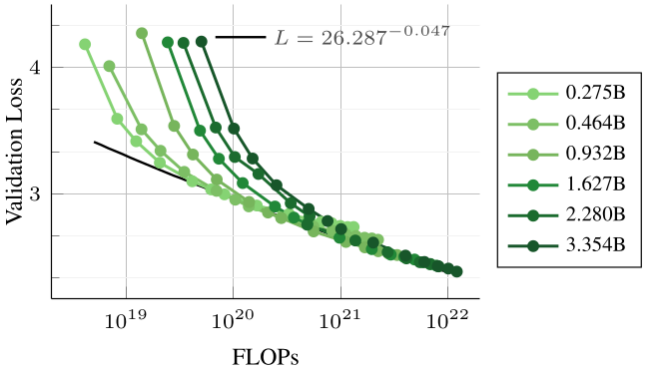


Figure: Scaling laws for sparse early-fusion NMMs

Interestingly, the exponent (-0.048) is close to that of dense NMMs (-0.049), indicating that both architectures scale similarly. However, the multiplicative constant is smaller for MoEs (27.086) compared to dense models (29.574), revealing lower loss.

Result 8: Modality-agnostic vs Modality-aware Routing

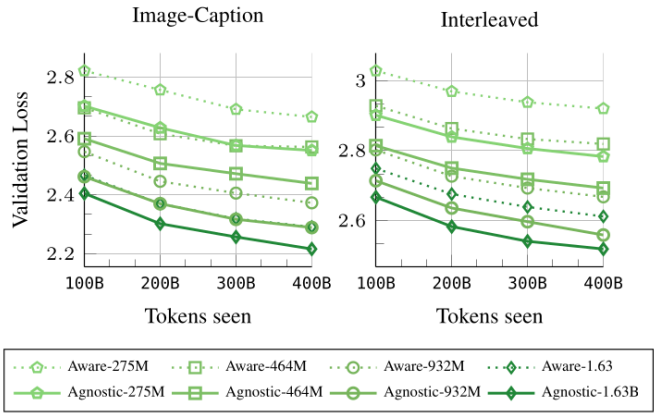
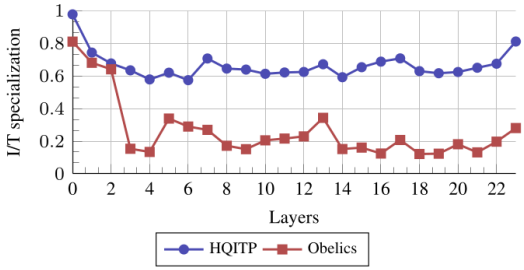


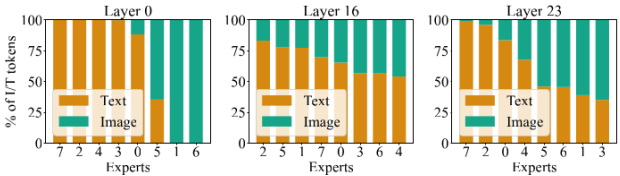
Figure: Modality-agnostic vs Modality-aware routing.

We find that modality-agnostic routing outperforms modality-aware routing across all model sizes, suggesting that allowing the model to learn its own routing strategy is more effective than imposing a fixed, modality-based routing scheme.

Result 9: Emergent Specialization



(a) Expert Specialization Score across layers.



(b) Expert Specialization Frequency.

- Experts automatically specialized into "Image" or "Text" experts. This happened most in **early** and **late** layers (processing raw inputs and final predictions).
- Middle layers remained more "multimodal" (shared). This behavior is intuitive, as they are expected to hold higher-level features that may generalize across modalities,

The emergence of both expert specialization and cross-modality sharing in our modality-agnostic MoE, suggests it may be a preferable approach compared to modality-aware sparsity

Result 10: Evaluation on downstream tasks with SFT

	Accuracy						CIDEr	
	AVG	VQAv2	TextVQA	OKVQA	GQA	VizWiz	COCO	TextCaps
Late-fusion	46.8	69.4	25.8	50.1	**65.8**	22.8	70.7	50.9
Early-fusion	47.6	69.3	28.1	**52.1**	65.4	23.2	**72.0**	53.8
Early-MoEs	**48.2**	**69.8**	**30.0**	**52.1**	65.4	**23.6**	69.6	**55.7**

Figure: Supervised fine-tuning on the LLaVA mixture. All models are native at 1.5B scale and pre-trained on 300B tokens.

Early fusion outperforms late fusion, and MoE outperform dense models. However, since the models are relatively small (1.5B scale), trained from scratch, and fine-tuned on a small dataset, the overall scores are lower than the current state of the art.

Limitations & Conclusion

Limitations

- Scaling law coefficients are broadly consistent across training mixtures, though a broader exploration is needed to validate this observation.
- While validation loss scales predictably with compute, the extent to which this correlates with downstream performance remains unclear and warrants further investigation.
- The accuracy of scaling law predictions improves with higher FLOPs, but their extrapolation to extreme model sizes is still an open question.

Future Work

- **High-Resolution Efficiency:** Early-fusion faces a massive increase in tokens at high resolutions. Future work must explore pooling techniques or context-extension methods rather than standard tokenization.
- **Advanced MoE Configurations:** Move beyond standard top-1 routing; explore different load-balancing strategies, routing granularity, and expert implementations to maximize specialization.
- **Downstream Task Correlation:** Definitively establish the link between pre-training validation loss (which scales predictably) and actual performance on benchmarks like VQA and Captioning.

Conclusion

Takeaways

1. **Architecture efficiency:** Late-Fusion offers no inherent advantage[cite: 13, 56]. Early-Fusion is significantly more efficient to train and performs better at lower compute budgets[cite: 14, 315].
2. **Predictable Scaling:** Native Multimodal Models (NMMs) follow strict scaling laws comparable to text-only LLMs, allowing for precise budget planning[cite: 58, 261].
3. **The MoE Advantage:** Sparse models outperform dense models at the same inference cost by implicitly learning modality-specific experts[cite: 73, 397].

Thank you!

cdimas@tuc.gr